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Computer Science Programming – Notes

Computer Science Programming Notes

1. Programming What is it?

Computer Programming involves computer instructions. Programming languages are used to create software, websites, applications and computer-based solutions.

2. Programming is logical thinking?

Programming involves logical thinking, problem solving, creativity and computer applications.

3. C Programming

C is a programming language. Programming basics, variables, operators, conditions, loops, functions and memory concepts are covered in C.

Basic C example

```
int a = 10;
int b = 20;
int sum = a + b;
```

4. Python

Python is a programming language. It is used for automation, data science, artificial intelligence, web development and applications.

Basic Python example

```
a = 10
b = 20
print(a + b)
```

5. Java

Java is an object-oriented programming language. Software, enterprise applications and applications are created using Java.

6. Variables

Variable is data. For example, age = 18, age is a variable.

7. Data Types

Common data types include Integer (whole numbers), Float (decimals), Character (single character), String (text), and Boolean (True/False).

8. Operators

Arithmetic operators: +, -, *, /, %

Comparison operators: ==, !=, >, <, >=, <=

Logical operators: AND, OR, NOT

9. If-Else Conditions

Condition determines if-else execution. If condition is True, execute code block; if False, execute alternative code block.

10. Loops

Loops allow repeating code. Common types: for loop (iterates over a range), while loop (runs until condition is false).

11. Functions

Function is a reusable block of code. Functions help organize code and avoid repetition.

12. Arrays

Array is a collection of values. Programming language defines array syntax and operations.

13. Object Oriented Programming (OOP)

OOP organizes program into objects and classes. Key concepts: Class, Object, Encapsulation, Inheritance, Polymorphism, Abstraction.

14. Basic Problem Solving

Programming problem involves understanding the problem, input/output, algorithm steps, code implementation, testing, and error handling.

Practice Questions

- 1 Programming 是什麼? 為什麼?
- 2 Programming languages 有什麼? 有什麼? 有什麼?
- 3 Variable 是什麼? 為什麼?
- 4 C 和 Python 有什麼? 有什麼? 有什麼?
- 5 Data type 是什麼? 為什麼?
- 6 If-Else 是什麼? 有什麼? 有什麼?
- 7 Loop 是什麼? 為什麼? 有什麼? 有什麼?
- 8 Function 是什麼? 為什麼?
- 9 OOP 是什麼? 為什麼?
- 10 Programming problem 是什麼? 有什麼? 有什麼? steps 是什麼?